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Analysis of Reading Assessment Data in the NAZ – Fall 2014

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Introduction and Summary

This report presents information results of reading assessments NAZ-enrolled scholars completed between the fall of 2013 and fall of 2014.

Specifically, this report addresses the following questions:

- How did scholars' reading scores change from fall 2013 to spring 2013?
- How did scholars' reading scores change from spring to fall 2014?
- How did scholars' reading change from fall 2013 to fall 2014?

In addition to presenting information on the average change in scholars' scores between these time points, the report provides information about those scholars assessed at multiple time points (fall 2013, spring 2014, and fall 2014), compared to those scholars only assessed in the fall of 2014.

Assessment Periods

Reading assessments were completed according to the following schedule:

- Fall 2013 – October to November 2013
- Spring 2014 – April to May 2014
- Fall 2014 – October to November 2014

Scholars were assessed at NAZ partner schools, NAZ partner Expanded Learning (ExL) Programs, NAZ's main offices, and, on rare occasions, in scholars' own homes. Details about the specific reading measures used are provided below.

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Authors' Note

This report was produced by the NAZ Internal Evaluation Team at the University of Minnesota; this team is directed by Scott McConnell, Ph.D., and includes Laura Potter, Joseph Demers, Lynn Edwards, and Kevin Overson. This work is supported by a contract with the Northside Achievement Zone, however all work is conducted independently and no official endorsement by the funder should be inferred.

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This report is available on the CEED@UROC NAZ Internal Evaluation website, and is available in alternate formats on request. For more information, please contact the authors at ceeduroc@umn.edu or 612-624-9381.

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Scholars Examined

This report looks specifically at scholars who were assessed both during the 2013-14 school year and the 2014-15 school year. A total of 245 scholars were assessed in the fall of 2013 and again in the spring of 2014. A total of 187 scholars who were assessed in spring 2014 were also assessed in fall 2014. Finally, a group of 91 scholars completed assessments at all three time points, allowing us to examine growth over one full calendar year for this small subset of scholars.

We also compared the scholars who were assessed during these two school years to those scholars who were only assessed in the fall of 2014. In examining differences between scholars who were assessed at multiple times and those assessed only in fall 2014, we considered what portion of the scholars assessed more than once were enrolled in a NAZ partner ExL program. Scholars' ExL programs were identified from one of the NAZ partner programs for each of the time periods as follows:

- 21st Century
- Ascension
- Beacons
- Breakthrough*
- Patchwork Quilt†
- PYC
- STA

**Summer and fall 2014 only*
†Spring 2014 only

Reading Measure

Scholars completed Curriculum-Based Measures of Reading (CBMRe) within the FAST online assessment system (www.fastforteachers.org). Each Scholar was asked to read three different passages aloud, each for 1 minute, while a staff member recorded the Scholar's words read correctly (WRC) and errors. Proficiency levels were determined by comparing the median (or middle score) of a scholar's three attempts to benchmark scores derived from studies of nationally representative samples. This assessment has been shown to effectively predict performance on more complex measures of reading proficiency, including standardized achievement tests.

In addition to the three standard grade level passages, during the fall 2014 assessments, scholars in grades two through six, who had been assessed the year before, also completed one passage from their grade from the previous year. Under this system, scholars in second grade in fall 2014 completed three second grade passages and one first grade passage. These additional passages were included to allow for comparisons of scholars' change in reading achievement from one year to the next.

Changes in scholars' reading achievement were calculated by subtracting their spring 2014 score from the previous grade passage they completed in fall 2014. In other words, for a scholar in second grade in fall 2014, that scholars' first grade score from spring 2014 was subtracted from her score on the first grade passage she completed in fall 2014.

We were able to use this system to examine changes in scholars who were in grades two through six in fall 2014. Since kindergarteners do not complete CBMRe, we could not compare score for scholars in first grade this year to their results from the previous year. In addition, since grades six, seven, and eight all use the same passages, scholars in grades seven and eight did not complete a passage from the previous grade; however, we could still compare their results from the fall of this year to the previous year.

Reading Achievement Results

Fall 2013 to Spring 2014 Change

We first examined the 245 scholars who were assessed in the fall and spring of the 2013-14 school year. The table below shows the number (N) of students in each grade that was assessed twice, the average or mean (M) score for each grade at each time point, and the standard deviation (SD) for that group of scores. The rightmost section shows the mean (M) change in these scholars' scores from fall to spring.

		Fall 2013		Spring 2014		Fall to Spring Change	
2013-14 Grade	**n**	**M**	**SD**	**M**	**SD**	**M**	**SD**
1	39	29.74	27.81	53.44	31.41	+23.69	18.70
2	64	63.61	38.83	84.17	40.88	+20.56	12.97
3	57	93.89	37.47	107.89	37.12	+14	15.73
4	44	104.45	40.59	117.57	36.29	+13.11	13.98
5	41	108.05	38.78	120.56	39.72	+12.51	15.35
TOTAL	245						

As these data indicate, scholars assessed in both the fall and the spring, on average, increased, with first and second grade students showing the largest average changes.

Spring to Fall 2014 Change

Scholar characteristics

We compared the students who were assessed both last year and this year with those scholars only assessed in fall 2014, to determine whether these two groups appeared different. A total of 405 scholars were assessed in grades two through eight in the fall of 2014. Of these scholars, 187 had been assessed the year before and completed an additional previous grade passage in fall 2014. On average, scholars who were assessed both last year and this year scored 20 points higher on CBMRe in the fall of 2014 than scholars who were only assessed in the fall.

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Looking more closely at the 187 scholars assessed both last year and this year, 82 (44%) were in a partner ExL program in the spring of 2014. Furthermore, 128 (68%) were in a partner ExL program during the summer of 2014, with an additional 44 (22%) in non-partner summer programs.

Spring to Fall Changes

The table below shows the average change in scholars' CBMRe scores from spring to fall 2014 for the 187 scholars assessed at both of these time points, including the number (n) of students, the mean (M) change in score, and the standard deviation (SD) of the scores for that group:

		Spring 2014		Fall 2014		Spring to Fall Change	
2013-14 Grade	**n**	**M**	**SD**	**M**	**SD**	**M**	**SD**
1	26	47.69	26.79	68.19	36.72	+20.5	15.59
2	36	78.67	36.57	88.14	44.07	+9.47	16.30
3	26	114.92	31.46	121.08	32.81	+6.15	12.56
4	15	126.47	24.99	142.73	31.91	+16.27	18.25
5	35	111.37	41.72	126.57	39.97	+15.2	24.74
6	29	143.34	38.52	151.97	34.80	+8.62	16.05
7	20	163.55	37.20	168.5	35.38	+4.95	18.07
TOTAL	187						

As these data indicate, the change from spring to fall varied considerably across scholars. The average change was highest among students moving from first to second grade and lowest among students moving from seventh to eighth grade. This is consistent with expectations that students make more modest increases in reading rate as they get older.

Year to Year Changes

The table below shows the average change in scholars' CBMRe scores from fall 2013 to fall 2014 for the 91 scholars assessed at both of these time points, including the number (n) of students, the mean (M) change in score, and the standard deviation (SD) of the scores for that group:

		Fall 2013		Fall 2014		Spring to Fall Change	
2013-14 Grade	**n**	**M**	**SD**	**M**	**SD**	**M**	**SD**
1	17	27.76	22.66	72.82	36.37	+45.06	24.13
2	26	65.62	35.12	96.27	44.55	+30.65	15.35
3	19	100.68	28.70	126.42	25.65	+25.74	16.39
4	12	113.42	33.18	145.33	32.00	+31.92	25.53
5	17	101.94	36.73	130.29	46.15	+28.35	15.59
TOTAL	91						

As these data indicate, this group of 91 scholars increased, on average, by at least 25 WRCM from last school year to the start of this school year. Students who moved from first grade into second grade, on average, demonstrated the largest increase.

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Summary

The increase in scholars' scores from spring to fall is notable, since this increase occurred over the summer. As noted in a previous report dated September 9, 2014, nationally, students typically demonstrate *decreased* reading achievement over the summer – a phenomenon known as "summer slide." The fact that NAZ scholars showed *increased* achievement is notable. Furthermore, more than two-thrds of the scholars (N=128) who were assessed both last year and this year were enrolled in NAZ Partner ExL programs over the summer, indicating that a considerable number of scholars continued their involvement with NAZ from the summer into the current school year. The ability to assess students at multiple time points is likely to provide information that will help inform instructional supports for those scholars. Lastly, at this point, we have data at three time points for 91 scholars, which is 22% of the total number of grade two to eight scholars assessed in fall 2014. Their average growth may not reflect the typical growth for the other 314 scholars assessed in fall 2014.

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